

- Q.28 Describe any five importance of surveying in civil engineering projects?
- Q.29 Convert the following WCB to QB:
 a) $142^{\circ}30'$ b) $223^{\circ}15'$ c) $60^{\circ}10'$
 d) $12^{\circ}45'$ e) $287^{\circ}15'$
- Q.30 Write any five advantages of plane table surveying.
- Q.31 Write any five disadvantages of chain surveying.
- Q.32 Find the correction for curvature and refraction for distance of
 a) 2933 meters b) 7 Km
- Q.33 Write a short note on orientation of plane table.
- Q.34 Describe the various types obstacle in chain surveying.
- Q.35 Write a short note on check leveling.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the mechanical methods for solving the three point problem
- Q.37 The following bearings were observed while traversing with a compass:

Line	FB	BB
AB	$80^{\circ}45'$	$26^{\circ}00'$
BC	$130^{\circ}30'$	$311^{\circ}35'$
CD	$240^{\circ}15'$	$60^{\circ}15'$
DA	$290^{\circ}30'$	$110^{\circ}10'$

Mention which stations were affected by local attraction and determine the corrected bearings.

- Q.38 The following staff readings were observed successively with a level, the instrument having been moved after 3rd, 6th and 8th reading: 2.228, 1.606, 0.988, 2.090, 2.864, 1.262, 0.602, 1.982, 1.044, 2.684 meters. Enter the above readings in a page of a level book and calculate the RL of points using height of instrument method, if the first reading was taken with a staff held on a bench mark of 432.384 m

No. of Printed Pages : 4 180733/170733/120733
 Roll No. /030733

3rd Sem / Civil, Brick Tech, Const Mgmt, Highway Engg.
Subject:- Surveying - I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A metallic tape is made of _____
 a) Steel b) Invar
 c) Linen d) Cloth and Wires
- Q.2 Local attraction in compass surveying may exist due to _____
 a) Presence of magnetic substance near the instrument
 b) Loss of magnetism of the needle
 c) Incorrect leveling of the magnetic needle
 d) Friction of needle at the pivot
- Q.3 If the fore bearing of a line AB is 285° , then the back bearing will be _____
 a) 75° b) 95°
 c) 105° d) 115°
- Q.4 The type of surveying which requires least office work is _____
 a) Trigonometrical leveling
 b) Theodolite surveying
 c) Tacheometry
 d) Plane table surveying
- Q.5 For a line AB _____
 a) The fore bearing of AB and back bearing of BA differ by 180°
 b) The fore bearing of AB and back bearing of AB differ by 180°

- c) Both (a) and (b) are correct
d) None is correct
- Q.6 Which of the following is an obstacle to chaining but not to ranging?
a) Building b) River
c) Hillock d) None of these
- Q.7 The correction for sag is _____
a) Always zero
b) Always additive
c) Always subtractive
d) sometimes additive and sometimes subtractive
- Q.8 Sensitiveness of a level tube is designated by _____
a) Radius of level tube
b) Length of bubble of level tube
c) Length of level tube
d) None of the above
- Q.9 The following sights are taken on a “Turning Point” _____
a) Back sight only
b) Foresight and Back Sight
c) Foresight only
d) Foresight and Intermediate sight
- Q.10 The instrument used for accurate centering in plane table survey is _____
a) Spirit level b) Alidade
c) Plumbing fork (U-fork)
d) Trough compass

SECTION-B

Note: Objective type questions. All questions are compulsory.
(10x1=10)

- Q.11 The instrument used for accurate centering in plane table survey is _____ (U-fork/ Alidade)

- Q.12 The size of plane table is _____ (600 mm x 750mm / 300 mm x 450 mm)
- Q.13 Axis of telescope is an imaginary line joining the centre of the eye piece and the optical centre of the _____ (object-glass / Leveling staff)
- Q.14 In an internal focusing type of telescope, the lens provided is (concave / convex)
- Q.15 Sensitiveness of a level tube is designated by _____ (Length of level tube / Radius of level tube)
- Q.16 The horizontal angle between the true meridian and magnetic meridian at a place is called _____ (magnetic bearing/Declination)
- Q.17 The length of a chain is measured from outside of one handle to _____ (Inside of other handle / Outside of other handle)
- Q.18 The correction for slope is required when the points of supports are not at the same level. (True/False)
- Q.19 An invar tape is made of an alloy of _____ (Nickel & Steel / Nickel & Brass)
- Q.20 The correction for sag is _____ (Always additive / Always subtractive)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Enlist the instrumental errors in leveling and describe any one of them.
- Q.22 Write a short note on orientation of plane table by trough compass.
- Q.23 Name the different classifications of surveying based on nature of field survey and describe any one of them.
- Q.24 Describe the process of temporary adjustment of a dumpy level.
- Q.25 Write five sources of error in compass work.
- Q.26 Name the different equipments used for plane table surveying.
- Q.27 Write five purposes of leveling.